

Skills Summary

Using Slopes to Prove Lines Parallel or Perpendicular

Use this reference while completing your worksheet or when studying.

Skill 1 — Parallel Lines Have Equal Slopes

Slope formula: for points (x_1, y_1) and (x_2, y_2) , $m = \frac{y_2 - y_1}{x_2 - x_1}$ — the rise over the run.

Parallel test: two distinct non-vertical lines are parallel *exactly when* $m_1 = m_2$. Equal slopes mean equal direction, so the lines never meet. Two vertical lines are parallel as well, even though neither has a slope.

Example: Line ℓ passes through $A(1, 1)$ and $B(5, 3)$. What is the slope of every line parallel to ℓ ?

Step 1. Parallel lines carry the same slope, so the whole question reduces to finding the slope of ℓ itself.

Step 2. $m = \frac{3 - 1}{5 - 1} = \frac{2}{4}$, reduced to lowest terms.
 $\Rightarrow m = \frac{1}{2}$

Try it: Line n passes through $C(-2, -1)$ and $D(2, 5)$. What is the slope of every line parallel to n ?

check: $\frac{2}{2} = 1$

Skill 2 — Slopes of Perpendicular Lines

Perpendicular test: two non-vertical lines are perpendicular *exactly when* $m_1 m_2 = -1$; equivalently, each slope is the **opposite reciprocal** of the other — flip the fraction and change the sign.

The exception: a vertical line has no slope at all, so the product rule cannot be used. A line perpendicular to a vertical line is horizontal, and its slope is 0.

Example: Find the slope of a line perpendicular to the line through $E(-1, 2)$ and $F(3, 4)$.

Step 1. Perpendicular means opposite reciprocal, so find the slope of $\overline{EF}$ first and then transform it.

Step 2. $m_{EF} = \frac{4 - 2}{3 - (-1)} = \frac{2}{4}$, which is $m_{EF} = \frac{1}{2}$; flipping it gives 2 and changing the sign gives the perpendicular slope. Check: $(\frac{1}{2})(-2) = -1$ ✓
 $\Rightarrow m = -2$

Try it: Find the slope of a line perpendicular to the line through $G(0, 5)$ and $H(2, -1)$.

check: $\frac{2}{1} = 2$ is the perpendicular slope, so $m = -\frac{1}{2}$

(!) Watch out: the opposite reciprocal changes *both* things. The perpendicular partner of $\frac{2}{3}$ is $-\frac{3}{2}$ — not $\frac{3}{2}$ (sign forgotten) and not $-\frac{2}{3}$ (fraction not flipped).

Skill 3 — Coordinate Proofs Using Slopes

How a coordinate proof works. Turn the claim into slopes, compute, compare, then say what the comparison proves:

- both pairs of opposite sides parallel $\Rightarrow$ parallelogram
- parallelogram with one pair of adjacent sides perpendicular $\Rightarrow$ rectangle
- midpoints come from $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$; lengths from the distance formula

Finish with a sentence naming the theorem you used — the arithmetic alone is not a proof.

Example: A triangle has third side $\overline{TU}$ from $T(0, 0)$ to $U(8, 4)$, and the midpoints of the other two sides are $M(1, 3)$ and $N(5, 5)$. Show $\overline{MN} \parallel \overline{TU}$.

Step 1. Parallel is a statement about direction, so compare the slope of the midsegment with the slope of the third side.

Step 2. $m_{MN} = \frac{5-3}{5-1} = \frac{2}{4}$, and $m_{TU} = \frac{4-0}{8-0} = \frac{4}{8}$ — the same number, so the segments are parallel.

$$\Rightarrow m = \frac{1}{2}$$

Try it: The midpoints of two sides of a triangle are $(2, 4)$ and $(7, 6)$. Find the slope of the midsegment joining them.

$$\frac{5}{2} = m \text{ check}$$